



Community

Your feed Latest expert opinions Groups

✓ Published

Edit

The Great Unlocking: How AI is De-Risking the Core Banking Overhaul

1 Like

19

01 December 2025 Be the first to comment

**Neeraj Aggarwal**Senior Project Manager
Infosys Limited**Location** **Followers** **Opinions**

Cranston 6 16



For the last decade, the banking industry has been engaged in a frantic race of "digital transformation." We have built slick mobile apps, intuitive neobank interfaces, and 24/7 chatbots. But for many financial institutions, this transformation is only skin deep.

Peel back the layers of a modern-looking banking app, and you will often find a "hollow core": a user interface that speaks the language of 2025 communicating with a mainframe back-end that speaks the language of 1985.

The Reality: It is estimated that nearly **90% of the world's top banks** still rely on legacy mainframes for their core processing—handling everything from transaction clearing to interest calculations.

For years, CIOs have faced a paralyzing dilemma. Replacing the core is likened to "changing the engine of a 747 while in mid-flight"—too risky, too expensive, and too slow. So, they patched it. They built layers around it. They waited.

But the waiting game is over. The arrival of **Generative AI** has fundamentally shifted the modernization equation. AI is no longer just a feature to add *to* the bank; it is the tool required to *rebuild* it. We are moving from the era of "maintenance" to the era of "automated modernization."

The Comprehension Gap: Solving the "Black Box" Problem

The single biggest hurdle to modernizing a legacy core is not writing new code; it is **understanding the old code**.



Many core banking systems run on millions of lines of **COBOL**, written by engineers who have long since retired. Documentation is often sparse, outdated, or non-existent. For a modern developer to rewrite a loan origination module, they must first spend months essentially acting as a digital archaeologist, trying to decipher business logic buried in decades of "spaghetti code."

This is where Large Language Models (LLMs) act as a force multiplier.

Unlike human developers, who can take weeks to analyze a single module, modern AI agents can ingest vast repositories of legacy code in minutes. They don't just "read" the code; they map the dependencies.

- **Automated Documentation:** AI tools can scan a 40-year-old COBOL script and generate a plain-English summary of exactly what business rules it executes.
- **Logic Extraction:** AI can extract the *intent* of the code—the actual business logic—and suggest how to architect that logic in a modern, cloud-native microservice.

By automating the "discovery" phase, AI reduces the timeline of modernization projects by **30-40%**. It turns the "Black Box" of the legacy core into a transparent, documented blueprint, giving CIOs the confidence to finally decouple their systems.

Surgical Precision: AI and the "Strangler Pattern"

If the "Big Bang" migration is a recipe for disaster, the industry standard alternative is the **Strangler Fig Pattern**.

Named after the fig tree that grows around a host tree and eventually replaces it, this architectural pattern involves gradually stripping capabilities away from the legacy monolith and moving them to modern



microservices one by one. Eventually, the legacy core is "strangled" out of existence.

While this approach reduces risk, identifying the "seams"—the safe places to cut into a million-line monolith without crashing the bank—is painstakingly slow. **AI acts as the scalpel for this surgery.**

- **AI Dependency Mapping:** Advanced AI models can visualize the entire call graph of a legacy system in 3D. They identify "clusters" of code with low coupling, highlighting the "low-hanging fruit" that can be moved first with minimal risk.
- **Generating the "Glue":** GenAI excels at writing **API wrappers** and "anti-corruption layers" needed to translate data between the old COBOL structures and modern JSON/RESTful standards. What used to take a team of integration engineers weeks can now be prototyped by AI in hours.
- **Parity Testing (The Safety Net):** The biggest fear in modernization is: *"Will the new system calculate the interest exactly the same way the old one did?"* AI can look at the legacy code and generate thousands of **test cases** to cover every edge case, ensuring mathematical **parity** before a single customer transaction is routed to the new system.

Conclusion: The Speed of Trust

Once the heavy lifting of migration is complete, the role of AI shifts. We are moving toward **Agentic AI** in banking.

In a modernized, cloud-native environment, AI agents won't just be passive chatbots. They will have the agency to interact with the core directly. Imagine an AI Agent that detects a reconciliation error, traces it back to a specific transaction anomaly, and auto-corrects the ledger entry—all without human intervention.



For decades, the banking sector has treated core modernization as a "necessary evil." AI has flipped the script. It has turned modernization into a manageable, predictable, and **accelerated** engineering task.

By using AI to decipher the past (legacy code), manage the transition (Strangler Pattern), and secure the future (automated testing), banks can finally break free from the constraints of the mainframe. The question for leadership is no longer "Is it safe to migrate?" but rather, **"Can we afford to remain frozen in time while our competitors run on AI?"**

About Author

Neeraj Aggarwal is a recognized modernization leader with deep expertise in AI, core banking, and legacy transformation. He advises financial institutions on building resilient, data-driven architectures and contributes thought leadership across industry forums and publications.

 1359

 Report

Share

 1 Like

 2

 19







Channels

[/artificial intelligence](#), [/retail banking](#)



Artificial Intelligence and Financial Services

Artificial Intelligence and Financial Services

 583 opinions  181 members  8 hours

Comments: (4)



A Finextra member

 20 January 2026

The emphasis on AI-assisted discovery, dependency mapping, and risk scoring is excellent. These are the exact capabilities needed to modernize safely without breaking mission-critical workflows. Very well articulated.

 **Like**

 **Report**



A Finextra member

 21 January 2026

This is one of the few pieces that captures the *real* modernization challenge facing banks today: AI cannot deliver meaningful value until the underlying core is structurally de-risked. Most industry discussions treat AI as a bolt-on capability, but this article correctly frames it as an architectural catalyst that forces institutions to confront decades of accumulated technical debt.

The distinction between “digital veneer” and “hollow core” is particularly important. Many banks have invested heavily in front-end digitization while leaving the mission-critical payment, lending, and settlement engines untouched. The result is an ecosystem where AI pilots look promising in isolation but fail to scale because

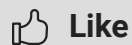
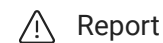


the foundational layers — lineage, data quality, dependency mapping, and workflow integrity — are not modernized. The article's framing of AI as a *risk-mitigation layer* rather than a risk amplifier is a refreshing and much-needed shift in thinking.

The emphasis on AI-assisted discovery, automated code intelligence, and dependency analysis is also spot-on. These capabilities are no longer experimental; they are becoming essential for safely unwinding legacy systems without jeopardizing regulatory compliance or operational stability. Banks that adopt this approach will be able to modernize with far greater confidence, especially in Tier-0 environments where downtime is not an option.

What stands out most is the practical, execution-oriented perspective. Instead of abstract strategy, the article outlines how AI can materially reduce modernization risk — from identifying brittle components to validating refactored logic to ensuring end-to-end workflow continuity. This is the kind of thinking that moves the industry forward, and it reflects a deep understanding of both the architectural and operational realities inside large financial institutions.

A very timely and insightful contribution. The industry needs more frameworks that connect AI innovation with modernization safety, and this article does that exceptionally well.

**Like****Report****Neeraj Aggarwal** Senior Project Manager at Infosys Limited**Author**

21 January 2026

Thank you for the thoughtful comment — you've captured the core tension that most banks are wrestling with today. The real breakthrough happens when institutions stop viewing AI as an "add-on" capability and start treating it as an analytical layer



that exposes structural weaknesses in the core. Once AI begins mapping lineage gaps, brittle workflows, and hidden dependencies, modernization stops being a blind rewrite and becomes a governed, evidence-driven transformation.

What we're seeing across large financial institutions is that AI doesn't just accelerate modernization — it *de-risks* it by giving teams far greater visibility into how legacy systems behave under stress. That visibility is what allows banks to modernize incrementally, validate refactored logic with confidence, and avoid the operational fragility that has historically made core transformation so risky.

The institutions that succeed are the ones that treat modernization as a dependency graph rather than a monolithic program. When AI-assisted discovery, code intelligence, and workflow validation are embedded early, the entire journey becomes more predictable, safer, and significantly more scalable.

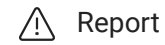
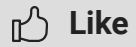
I appreciate you raising this point — it's exactly the kind of discussion the industry needs as we move from AI experimentation to AI-enabled modernization at scale.

**Like****Report****A Finextra member**

25 January 2026

This piece captures a key shift in the industry: AI isn't just accelerating modernization—it's actively reducing the risk that has held banks back for years. By improving code visibility, predicting downstream impacts, and strengthening validation, AI is turning core transformation from a high-uncertainty effort into a more controlled, data-driven process. Banks that treat AI as an architectural enabler—not just a productivity tool—will modernize faster and with far greater confidence.





B *I* U  

Have your say...

☒ As myself ☐ Anonymously

 **Post comment**



Neeraj Aggarwal
Senior Project Manager



Infosys Limited

Member since	Location	Followers	Following	Opinions
25 Nov 2025	Cranston	6	8	16

The Coming Failure of U.S. Digital Payments

 09 March

The Next Frontier in Payments: Why Modernization Must Move Beyond Speed n Intelligent Orchestration

 03 March

AI-Ready Payments: Why Legacy Payment Architectures Cannot Support Safe, Explainable, and Real-Time

 23 February

How AI and IoT Is Transforming Digital Payment Systems: A Modernization Blueprint for Banks

 20 February

The Vanishing Assembler Workforce: A Hidden Risk to Global Financial Stability

 16 February

[See all Opinions from Neeraj](#)

More expert opinions



Vitaliy Abayev Chief Operational Officer at DCM
Deposits are shifting. What's driving it and what banks do next



 8 hours

Anthony Villis Managing Director at First Wealth
Planning your exit? Here are five things you need to consider

 8 hours

Chetan Channe Marketing Head at Kaiso Research and Consulting
The Pilot Graveyard: Why Bank AI Projects Stall Between Proof of Concept and Production

 8 hours  1

Avik Nandi PRODUCT MANAGER at WIPRO LIMITED
Stablecoins to Intelligent Payment Routing: Building an AI Payment Orchestration Layer (Prototype)

 9 hours[See all opinions](#)

Finextra®

Finextra

[About](#)

Community

[Rules](#)

News

[Guidance](#)[Contact the news desk](#)

Sales

[Media pack](#)

Get involved

[Finextra Live@](#)[Webinars](#)[Finextra TV](#)[Research](#)[Finextra.jobs](#)

Events

[Sustainable Finance Live](#)[NextGen Nordics](#)[EBAday](#)

[Contact the community team](#)

[Contact the sales team](#)

Finextra Pro

Members

 [Manage your account](#)

Download Finextra Pro

Discover **Finextra** PRO

Follow



Download Finextra News



Download
Finextra News



DOWNLOAD
Finextra News

© Finextra Research 2026

[Terms of use](#)

[Privacy Policy](#)

[Cookie Centre](#)